Teardown Benchmark Study

Powertrain cooling pack - radiator, charge air cooler and fan drive
Subject vehicle against two construction-duty benchmarks

Document number	CVOEM/VAVE/TD-BM/2024/014
Revision	Rev 1.0
Issue date	August 2024
Prepared by	VAVE Benchmarking Cell with Thermal Systems
Reviewed by	Chief Engineer - Powertrain Installation
Test condition	Sustained 6% grade, 40 t GCW, 38 C ambient
Vehicle class	HCV construction duty
Distribution	VAVE Committee; Thermal Systems; Powertrain Installation

Summary

The cooling packs of two construction-duty benchmarks were compared against the subject vehicle, with coolant temperature measured on a sustained grade at high ambient. The subject vehicle reaches the alarm limit after about twelve minutes of sustained climb. An increased-margin radiator core, about 15 per cent larger frontal area, paired with an electronically controlled viscous fan, is benchmarked as the measure to remove the excursion; a thermal simulation and hot-climate durability trial are still pending.

Contents

1	Scope and test method	2
2	Cooling pack construction	4
3	Powertrain cooling pack	6
4	Opportunities and conclusions	8

SYNTHETIC ILLUSTRATIVE - NOT A CVOEM RECORD

1 Scope and test method

This study covers the radiator, charge air cooler, condenser, fan, fan drive and shroud, and the air path from the grille to the fan. Engine-side coolant circuit components are excluded.

1.1 Thermal test

Each vehicle was driven at 40 tonnes gross combination weight up a sustained 6 per cent grade at 38 degrees Celsius ambient, in the gear giving maximum sustained power, until coolant temperature stabilised or the alarm limit of 105 degrees Celsius was reached. Coolant temperature, charge air temperature and fan duty were logged at 1 Hz.

1.2 Why this condition

Construction-duty vehicles spend a disproportionate share of their operating life at high load and low road speed, where ram air is minimal and the fan carries the full cooling demand. A cooling pack sized on a highway duty cycle is under-specified for this use, and the resulting excursions appear as derate complaints rather than as failures.

> Method note. The test is a sustained-grade condition, not a peak ambient soak. A pack that passes a soak test can still reach the alarm limit on a grade, because the limiting factor is air mass flow through the pack rather than the coolant volume.

SYNTHETIC / ILLUSTRATIVE - NOT A CVOEM RECORD

1.3 Measured result

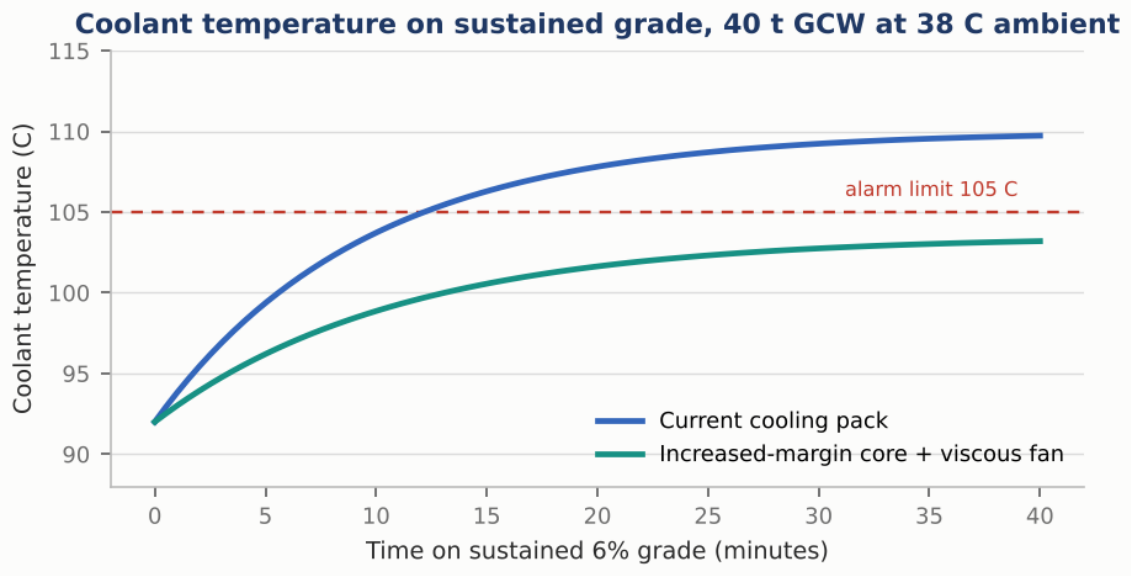


Figure 1 - Coolant temperature against time on a sustained 6 per cent grade at 40 t GCW and 38 C ambient. The current pack crosses the alarm limit at about twelve minutes.

Measurement	Subject vehicle	Benchmark F	Benchmark G
Stabilised coolant temp (C)	109.8 (alarm)	101.4	103.2
Time to alarm limit (min)	12.1	Not reached	Not reached
Charge air outlet temp (C)	68.4	58.2	61.0
Fan duty at stabilisation (%)	100	82	88
Measured fan power (kW)	18.4	14.1	15.8

Finding 1.1. The subject vehicle reaches the alarm limit after about twelve minutes of sustained climb; neither benchmark reaches it. The fan is at 100 per cent duty at stabilisation, so no reserve remains.

SYNTHETIC / ILLUSTRATIVE - generated for an academic dissertation prototype. Not a genuine CVOEM record.

2 Cooling pack construction

2.1 Core comparison

Attribute	Subject	Benchmark F	Benchmark G
Radiator frontal area (sq m)	0.71	0.82	0.79
Core depth (mm)	48	52	56
Fin density (fins/inch)	12	11	11
Charge air cooler area (sq m)	0.44	0.52	0.49
Condenser area (sq m)	0.31	0.31	0.33
Pack mass (kg)	62.4	71.8	74.2

Both benchmarks use a larger frontal area with a lower fin density. Lower fin density reduces heat transfer per unit area but also reduces air-side pressure drop, which raises mass flow for a given fan power. At high fan duty the second effect dominates.

Finding 2.1. The subject pack is optimised for heat transfer per unit area rather than for air mass flow. At 100 per cent fan duty this is the wrong optimisation, because the pack is air-limited rather than area-limited.

SYNTHETIC / ILLUSTRATIVE - NOT A CVOEM RE

2.2 Fan and drive

The subject vehicle uses a thermostatic viscous fan drive that engages on under-bonnet air temperature. Both benchmarks use an electronically controlled viscous drive commanded on coolant and charge air temperature together.

Attribute	Subject	Benchmark F	Benchmark G
Fan drive type	Thermostatic viscous	Electronic viscous	Electronic viscous
Control input	Under-bonnet air temp	Coolant and charge air	Coolant, charge air, AC
Fan diameter (mm)	710	750	750
Blade count	9	11	9
Engagement lag (s)	42	4	6

The 42 second engagement lag on the thermostatic drive is significant on a grade: by the time under-bonnet air temperature has risen enough to engage the drive, coolant temperature has already climbed. The electronic drives respond to the coolant directly and engage within a few seconds.

Finding 2.2. Engagement lag, not fan capacity, is the first-order difference. A proposal that increases fan diameter without addressing the control input would add parasitic power without removing the excursion.

SYNTHETIC ILLUSTRATIVE - NOT A CVOEM RECORD

3 Powertrain cooling pack

3.1 Benchmarked measure

The measure benchmarked on both comparison vehicles is an increased-margin radiator core, about 15 per cent larger frontal area, paired with an electronically controlled viscous fan. It is intended to reduce sustained-grade coolant-temperature excursions. A thermal-simulation and hot-climate durability trial is still pending.

Element	Current	Proposed	Change
Radiator frontal area (sq m)	0.71	0.82	+15.5%
Fin density (fins/inch)	12	11	-8.3%
Fan drive	Thermostatic viscous	Electronic viscous	-
Engagement lag (s)	42	5	-88%
Pack mass (kg)	62.4	70.2	+7.8 kg
Est. pack cost (INR)	34,200	46,800	+12,600

Finding 3.1. An increased-margin radiator core, about 15 per cent larger frontal area, paired with an electronically controlled viscous fan, is benchmarked as the measure to remove the sustained-grade excursion. Thermal simulation and a hot-climate durability trial are pending.

SYNTHETIC ILLUSTRATIVE - NOT A CVOEM

3.2 Expected result

On the measured air-side characteristics of the benchmark cores and the subject vehicle's heat rejection, the proposed pack is estimated to stabilise at about 103 degrees Celsius on the test grade, 2 degrees below the alarm limit. That is a smaller margin than either benchmark achieves and should be confirmed by simulation before commitment.

Caution. The estimated 2 degree margin is within the uncertainty of the calculation. This measure should not be released on this evidence alone; the pending simulation is a gate, not a formality.

3.3 Packaging

A 15 per cent larger frontal area requires 46 mm of additional pack height, which is available within the current grille aperture but reduces clearance to the bonnet latch cross-tube to 11 mm. Both benchmarks package a larger pack in a comparable aperture by moving the condenser below rather than in front of the radiator.

3.4 Secondary benefit

Measured fan power falls from 18.4 kW to an estimated 14.6 kW because the electronic drive runs at part duty for most of the cycle rather than cycling between zero and full. On a construction duty cycle this is worth an estimated 1.8 per cent in fuel consumption, which is not currently counted in the business case.

SYNTHETIC
ILLUSTRATIVE - NOT A CVOEM RECORD

4 Opportunities and conclusions

Ref	Opportunity	Cost (INR)	Class	Status
TD-14-01	Electronic viscous fan drive alone	+6,400	Running change	Recommended
TD-14-02	Increased-margin radiator core alone	+6,200	Platform change	Not alone
TD-14-03	Both, as a package	+12,600	Platform change	Pending simulation
TD-14-04	Condenser relocation below radiator	+2,100	Platform change	Enables TD-14-02

- The subject vehicle reaches the coolant alarm limit after about twelve minutes on a sustained 6 per cent grade at 40 t and 38 C.
- Engagement lag of 42 seconds on the thermostatic fan drive is the first-order cause, not core size.
- TD-14-01, the fan drive alone, is a running change and captures most of the benefit. It should be released first and measured.
- The full package is estimated to stabilise 2 degrees below the alarm limit, which is inside the calculation uncertainty and must be confirmed by simulation.
- The 1.8 per cent fuel benefit from part-duty fan operation is not currently in the business case and would materially improve it.

Recommendation. Release TD-14-01 and re-measure the grade test before committing to the core change. If the fan drive alone removes the excursion, TD-14-02 and TD-14-04 are unnecessary and INR 8,300 per vehicle is avoided.

SYNTHETIC ILLUSTRATIVE - NOT A GENUINE CVOEM RECORD